Ship Detection in SAR Images via Enhanced Nonnegative Sparse Locality-Representation of Fisher Vectors

Xueqian Wang, *Member, IEEE*, Gang Li, *Senior Member, IEEE*, Antonio Plaza, *Fellow, IEEE*, and You He

***Abstract*— As a powerful coding strategy for superpixels in synthetic aperture radar (SAR) images, Fisher vector (FV) lies in a low-dimensional subspace and can be sparsely represented as a linear combination of training samples. The existing ship detection methods based on FVs often consider the Euclidean distances between target FVs and clutter FVs, where the subspace features of FVs are generally not exploited. In this article, we propose a new ship detection algorithm based on nonnegative sparse locality-representation (NSLR) to exploit the subspace features of FVs. The proposed NSLR method is based on the assumption that FVs of superpixels in SAR images are sparsely represented by the dictionary of background sea clutter only under a null hypothesis. In addition, we propose two FV-based filters to enhance the robustness of our newly developed NSLR to heterogeneous sea clutter environments by further exploiting the intrinsic features of ship targets in terms of intensity and spatiality. The experimental results based on Gaofen-3 SAR images demonstrate that the proposed NSLR detection method provides higher target-to-clutter contrast and achieves better detection performance than other commonly used ship detection algorithms.**

***Index Terms*— Fisher vectors (FVs), ship detection, sparse representation (SR), superpixels, synthetic aperture radar (SAR).**

I. Introduction

RECENTLY, the problem of ship target detection has aroused much interest due to its key role in ship traffic service, sustainable fishery, maritime security, natural resource extraction, and so on [1]–[13]. As an active remote sensing technique, synthetic aperture radar (SAR) can generate data with high resolution over large areas in all weather conditions and has been an important tool for ship target detection [14]–[24].

A key problem for ship target detection is how to discriminate the ship targets from the background sea clutter. Generally, ship target detection methods for SAR images are based on the concept of constant false alarm rate (CFAR) [13]–[17]. CFAR-based ship detectors adaptively control the decision threshold according to the *a priori* probability density function (PDF) of the sea clutter and the expected false alarm rate [13]. In [3], the search for unidentified maritime object (SUMO)-based CFAR ship detector is introduced, which performs well on a wide range of SAR image modes and imaging resolutions. Liu *et al.* [4] proposed a CFAR detector based on truncated statistics to achieve improved detection performance in high-resolution SAR images with crowded ship targets. In [5], the performance of three CFAR detectors is compared based on a large data set containing thousands of ship targets to show their advantages and disadvantages, respectively. In addition to CFAR-based detectors, some detectors are also developed by further exploiting the characteristics of the ship targets and the background clutter [8], [12], [24]. Leng *et al.* [8] show the capability of kurtosis analysis in distinguishing the ship targets and the sea clutter and further design the corresponding ship detector. In [12], a ship detector based on the 2-D singularity power spectrum of SAR images is proposed to enhance ship detection performance in extremely low signal-to-clutter ratio (SCR) cases. In [24], a Bayesian model is established to decompose the SAR image as a sparse ship component plus sea clutter, and then the task of ship target detection is performed by separating a sparse signal from the SAR image. Recently, deep learning has been investigated as a mechanism to perform ship detection in SAR images [1], [7], [10]. This strategy can automatically extract features of SAR images but requires a large amount of labeled data. It is worth mentioning that most of the above ship detection algorithms focus on the pixel-level features of SAR images. In practice, a ship target often occupies a group of neighboring pixels instead of isolated pixel points, especially in high-resolution SAR images. However, most pixel-level detectors cannot accurately delineate the boundaries of ship targets and decide whether a group of neighboring pixels belong to the target area or the background clutter [18].

Superpixels exploit contextual information in high-resolution images and can better delineate the boundaries of targets [25]. Besides, target features extracted from superpixels

Manuscript received June 29, 2020; revised September 26, 2020 and November 1, 2020; accepted November 30, 2020. Date of publication January 8, 2021; date of current version October 26, 2021. This work was supported in part by the National Natural Science Foundation of China under Grant 61790551 and Grant 61925106, in part by Post-Doctoral Innovative Talent Support Program under Grant BX20200195, in part by China Postdoctoral Science Foundation under Grant 2020M680561, in part by Shuimu Tsinghua Scholar Program, and in part by FEDER and Junta de Extremadura under Grant GR18060. *(Corresponding author: Gang Li.)*

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Digital Object Identifier 10.1109/TGRS.2020.3042506

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can reduce the effect of speckle noise in SAR images [26]. Superpixels are also helpful in reducing the complexity of subsequent image processing tasks [27]. A popular method for the superpixel segmentation of SAR images is the simple linear iterative clustering (SLIC) [25]. Based on the intensity and spatial features of pixels, SLIC can generate superpixels with low complexity and good boundary delineation. In recent years, a few variants of SLIC have also been developed to further enhance the segmentation performance in SAR images [11], [27].

Owing to the superiority of superpixels, several existing works have been devoted to superpixel-based ship detection in SAR images [9], [18]–[20], [28]–[32]. In [18] and [29], the gap between the average intensities of the target superpixel and the neighboring clutter superpixels is used to design superpixel-based ship detectors. Pappas *et al.* [19] use superpixels (with a much larger size than that of the ship targets) instead of rectangular sliding windows to define guard bands in CFAR, thus reducing the number of false alarms. In [20] and [30], the information entropy of intensities in the SAR image is exploited to design superpixel-based ship detectors. In [31], a superpixel-based Bhattacharyya distance is used to discriminate the ship targets from the background clutter.

Recently, Fisher vector (FV) has been introduced as an alternative superpixel-based feature for ship target detection in SAR images [9], [28]. The FV of a superpixel is the gradient vector of the log-likelihood function with respect to the weights, mean values, and standard deviations in the global Gaussian mixture model (GMM). Therefore, the zero-order, first-order, and second-order characteristics of superpixels are captured simultaneously by FVs [33], [34]. In [28], a ship detector based on labeled FVs is proposed which achieves good detection performance (with a very small number of training samples). Wang *et al.* [9] exploit the contextual contrast of FVs and propose an improved FV-based ship detector. It is worth mentioning that the FV lies in the low-dimensional subspace spanned by the corresponding training samples from the same class [35], [36]. However, the existing ship detectors based on FVs [9], [28] mainly focus on the (weighted) Euclidean distances between the target FVs and clutter FVs and do not exploit the subspace features of FVs.

In the past decade, sparse representation (SR) has emerged as a powerful tool in exploiting subspace features and has been widely used in the context of face recognition [37], iris recognition [38], and hyperspectral image processing [39]–[42]. SR is based on the concept that a given test sample can be represented as the linear combination of training samples from the same class, where only a small number of representation coefficients are nonzero [37]. Unknown representation coefficients can be estimated by using sparse signal recovery [43]. Then, representation residuals are calculated to measure the distances between test samples and the subspace spanned by training samples. The main advantage of SR is that it can reduce the effect of the variability of training samples belonging to the same class [37]. Sparse locality-representation (SLR)[1] is an improved version of SR, which adds a locality constraint into SR and aims at representing a test sample via a few representative nearby training samples [44], [45]. Note that both the SR and SLR are not concerned with the signs of representation coefficients. The fact that similar (training) samples can cancel each other out using the subtraction operation is contrary to the notion of combining them to represent a test sample. To address this problem, the model of nonnegative SLR (NSLR) is formulated in [46]–[48], where a nonnegative constraint is imposed on the representation coefficients. In [46]–[48], it has been demonstrated that NSLR provides better representation capabilities for test samples than both the SR and SLR. To our best knowledge, NSLR has not yet been considered for the problem of ship detection in SAR images.

The main innovative contributions of this article can be summarized as follows.

1) We propose a new superpixel-based ship detector to exploit subspace features of FVs via NSLR. First, superpixel segmentation is performed on the SAR image, and then FV is calculated as the coding vector of each superpixel in the SAR image. For a test superpixel, a dual-window strategy is used to construct the background clutter dictionary (BCD), which contains FVs of superpixels located at the dual-window area of the test superpixel. Next, the representation residual of NSLR is calculated as the distance between the FV of the test superpixel and the subspace spanned by the BCD. This distance is used to discriminate between the ship targets and the background clutter in SAR images.
2) Two filters are designed to further enhance the performance of the proposed NSLR detector. NSLR formulates the problem of ship target detection as an anomaly detection one. Here, "anomaly" means that we know that there are differences between the ship targets and the background clutter, but we do not have *a priori* information on how to characterize the ship targets. In practice, the ship targets exhibit two inherent features in terms of spatiality and intensity, that is, they occupy far fewer pixels than the background sea clutter and appear as bright parts of the SAR image. Therefore, we propose two filters based on the zero-order and first-order information in FVs, respectively, to further exploit the spatiality and intensity features of the ship targets. These two filters are performed on the representation residuals obtained using NSLR to improve the capability of discrimination between the ship targets and the sea clutter.

An overview of the proposed approach is provided in Fig. 1. Compared with the existing FV-based ship detectors [9], [28], subspace features of FVs are newly considered in the proposed approach. Compared with the original NSLR model for anomaly detection in [46]–[48], target features are further exploited in the proposed approach. Our experimental results demonstrate that the proposed NSLR detection method outperforms the state-of-the-art approaches, thanks to the subspace features exploited by NSLR and the target enhancement capability of the proposed filters.

[1]SLR is also termed by locality-constrained linear coding [44].

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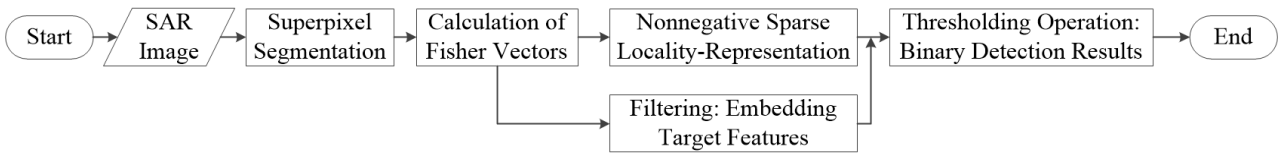


Fig. 1. Flowchart of the proposed NSLR detector for ship detection in SAR images.

The remainder of this article is structured as follows. Section II formulates the proposed NSLR detector. In Section III, we compare the detection performance and complexity of the proposed NSLR detector with those of other commonly used detectors using SAR images collected by the Gaofen-3 satellite. Section IV concludes this article with some remarks and hints at plausible future research lines.

II. Proposed Ship Detection Algorithm via NSLR

In this section, we first propose a new detection method based on NSLR that exploits the subspace features of FVs. In NSLR (and also the FV-based ship detectors in [9]), the ship targets are considered as anomalies in the SAR image, that is, the intrinsic features of the ship targets in terms of intensity and spatiality are not used. Therefore, two filters are also designed using FVs to embed the intrinsic target features into the NSLR, and thus enhance its detection performance. The proposed NSLR method is based on five steps: 1) superpixel segmentation; 2) calculation of FVs for superpixels; 3) NSLR of FVs; 4) filtering the representation residual map obtained by NSLR; and 5) thresholding operation. The details of the proposed method are presented below.

A. Step 1: Superpixel Segmentation

The goal of superpixel segmentation is to generate locally coherent image regions to reveal the boundary information of the ship targets. In this section, we use the SLIC algorithm in [25] to achieve the superpixel segmentation of SAR images. Other superpixel segmentation methods, such as those in [11] and [18], can also be used in this step. Here, we consider SLIC because of its computational simplicity and good segmentation performance [25].

The number of pixels in the SAR image is denoted by I. The expected superpixel size in the image is S, that is, the number of pixels in a superpixel is approximately $S \times S$. The distance measure between the pixel i and the pixel j is defined as [28]

$$D(i,j)=\sqrt{\left(\frac{\tau}{S}\right)^2\left[\left(x_i-x_j\right)^2+\left(y_i-y_j\right)^2\right]+\left(\delta_i-\delta_j\right)^2} \tag{1}$$

where $i, j \in \{1, 2, \ldots, I\}$, (δ_i, x_i, y_i) and (δ_j, x_j, y_j) represent the intensity values of the horizontal and vertical coordinates of the pixel i and the pixel j, respectively, and τ is used to balance the spatiality and the intensity dissimilarities. According to the distance in (1), the SLIC locally clusters pixels into superpixels. Next, we briefly summarize the iterative process of SLIC for completeness.

Step 1a: Initialize the centers of superpixels with regular grid interval S.

Step 1b: Associate each pixel with its nearest superpixel center using the distance measure defined in (1). The search scope for each superpixel center is $2S \times 2S$.

Step 1c: Update each superpixel center via the mean coordinate of pixels corresponding to it.

Step 1d: Repeat steps 1b and 1c until the number of iterations exceeds the predefined count *Iter*.

Step 1e: Postprocessing to enforce local connectivity.

We refer interested readers to [25] (or [28]) for more details about SLIC.

The number of superpixels generated by SLIC is denoted by P. The pth superpixel is represented by $\boldsymbol{\delta}_p \triangleq \{\delta_{p,1}, \delta_{p,2}, \ldots, \delta_{p,L_p}\}$, where L_p is the number of pixels contained by the pth superpixel, and $p = 1, 2, \ldots, P$. In Section III, the setting of hyper-parameters in SLIC will be discussed.

B. Step 2: Calculation of FVs

In this section, we calculate FVs as coding vectors of superpixels in the SAR image. First, GMM is constructed to model the universe SAR image. Then, FVs of superpixels are derived (based on GMM) to measure the deviation between the current superpixel and the global image.

GMM is known as a flexible model, which is the linear combination of N_{G} ($N_{\mathrm{G}} > 1$) Gaussian components and can approximate any continuous distribution with arbitrary precision [49]. The PDF of GMM is

$$g(\delta)=\sum_{n=1}^{N_{\mathrm{G}}}\omega_n\mathcal{N}\left(\delta;\mu_n,\sigma_n^2\right) \tag{2}$$

where ω_n, μ_n, and σ_n are the mixture weight, mean value, and standard deviation of the nth Gaussian function $\mathcal{N}(\delta; \mu_n, \sigma_n^2)$, respectively,

$$\mathcal{N}\left(\delta;\mu_n,\sigma_n^2\right)\triangleq\frac{1}{\sqrt{2\pi}\sigma_n}\exp\left[-(\delta-\mu_n)^2/2\sigma_n^2\right]$$

and $\sum_{n=1}^{N_{\mathrm{G}}}\omega_n = 1$. For ease of notation, let

$$\boldsymbol{\Lambda}\triangleq\{\omega_n,\mu_n,\sigma_n,n=1,2,\ldots,N_{\mathrm{G}}\} \tag{3}$$

denote the parameter set involved by GMM, which can be pre-estimated based on the maximum likelihood estimation (MLE) methods [50].

The FV of the pth superpixel is calculated as the (normalized) gradient vector of the log-likelihood function with respect to the parameter set $\boldsymbol{\Lambda}$ [33]

$$\mathbf{f}_p=\boldsymbol{\Phi}\sum_{l=1}^{L_p}\nabla_{\boldsymbol{\Lambda}}\log g\left(\delta_{p,l}\right),\quad\forall p \tag{4}$$

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where $\nabla_{\mathbf{\Lambda}}$ represents the gradient operation with respect to $\mathbf{\Lambda}$, $\mathbf{\Phi}$ denotes the Cholesky decomposition of $\mathbf{F}^{-1}$,

$$\mathbf{F} = \mathrm{E}_{\delta_p}\left\{\sum_{l=1}^{L_p} \nabla_{\mathbf{\Lambda}} \log g\left(\delta_{p,l}\right)\left[\sum_{l=1}^{L_p} \nabla_{\mathbf{\Lambda}} \log g\left(\delta_{p,l}\right)\right]^T\right\}$$

is the Fisher matrix, $\mathrm{E}_{\delta_p}(\cdot)$ is the statistical expectation with respect to $\boldsymbol{\delta}_p$, and $(\cdot)^T$ denotes the transpose operation. In the following, we provide the detailed formulation of the FV $\mathbf{f}_p \in \mathbb{R}^{3N_\mathrm{G}\times 1}$:

$$\mathbf{f}_p = \left[f_{p,\omega_1}, f_{p,\omega_2}, \ldots, f_{p,\omega_{N_\mathrm{G}}}, f_{p,\mu_1}, f_{p,\mu_2}, \ldots, f_{p,\mu_{N_\mathrm{G}}}, f_{p,\sigma_1}, f_{p,\sigma_2}, \ldots, f_{p,\sigma_{N_\mathrm{G}}}\right]^T \quad (5)$$

where

$$f_{p,\omega_n} = \frac{1}{L_p\sqrt{\beta_n}}\sum_{l=1}^{L_p}\left(\phi_{p,l}^{(n)} - \beta_n\right) \quad (6)$$

$$f_{p,\mu_n} = \frac{1}{L_p\sqrt{\beta_n}}\sum_{l=1}^{L_p}\phi_{p,l}^{(n)}\left(\frac{\delta_{p,l}-\mu_n}{\sigma_n}\right) \quad (7)$$

$$f_{p,\sigma_n} = \frac{1}{L_p\sqrt{2\beta_n}}\sum_{l=1}^{L_p}\phi_{p,l}^{(n)}\left[\frac{\left(\delta_{p,l}-\mu_n\right)^2}{\sigma_n^2} - 1\right] \quad (8)$$

and $\beta_n = (\exp(\omega_n)/\sum_{\tilde{n}=1}^{N_\mathrm{G}}\exp(\omega_{\tilde{n}}))$, $\phi_{p,l}^{(n)} = (\omega_n\mathcal{N}(\delta_{p,l};\mu_n,\sigma_n^2)/\sum_{\tilde{n}=1}^{N_\mathrm{G}}\omega_{\tilde{n}}\mathcal{N}(\delta_{p,l};\mu_{\tilde{n}},\sigma_{\tilde{n}}^2))$, for $p = 1, 2, \ldots, P$, $n = 1, 2, \ldots, N_\mathrm{G}$, $l = 1, 2, \ldots, L_p$. The elements in (6)–(8) capture the zero-order, first-order, and second-order characteristics of the pth superpixel, respectively [9], [28], [33]. Note that the dimension of FV in (5) is determined by the number of components in the GMM and is independent with regard to the size of superpixels. After (4)–(8), the signed square-rooting is used to avoid FVs that are close to null [51]

$$\mathbf{f}_p \leftarrow \mathrm{sign}\left(\mathbf{f}_p\right)|\mathbf{f}_p|^{1/2}, \quad \forall p \quad (9)$$

where $\mathrm{sign}(\cdot)$ and $|\cdot|^{1/2}$ are the element-wise operations and $\mathrm{sign}(\cdot)$ is 1 if the input is positive and -1 otherwise. Finally, the FVs of all the superpixels are self-l_2-normalized [52] for fair comparison in the subsequent NSLR (see Section II-C).

C. Step 3: NSLR of FVs

NSLR is formulated based on the fact that the FVs of the clutter superpixels in a local region lie in a limited subspace belonging to the redundant BCD. If the test superpixel is an "anomalous" ship target, the active bases in BCD show significant dissimilarity with regard to the FV of the anomaly test superpixel. On the other hand, when the test superpixel is a normal one (like those in the background clutter), its FV can be exactly represented using active bases in BCD. By comparing the representation residual with the decision threshold, the NSLR can be considered as an unsupervised subspace detector to check whether there is a salient difference between the test superpixel and its local background. In this section, we introduce our NSLR to distinguish the ship targets from the background sea clutter. NSLR mainly contains three substeps, that is, construction of BCD, calculation of SR coefficients, and calculation of representation residuals.

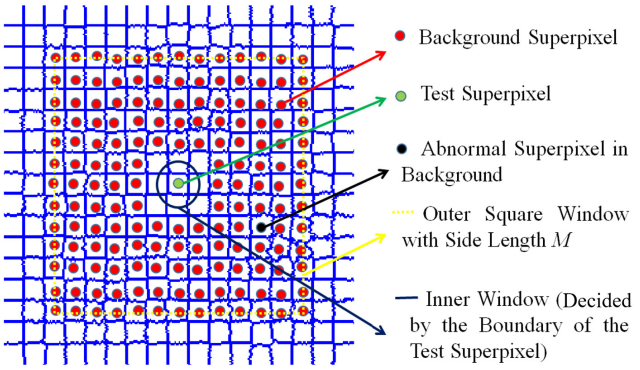


Fig. 2. Sketch of the dual-window strategy adopted for the construction of the BCD. The blue lines represent the boundaries of superpixels. The superpixels with red circles denote the background superpixels of the test superpixel with green circle. The superpixel with the black circle is the potential outlier superpixel in the background clutter, e.g., a target superpixel. The yellow dotted line represents the outer square window, which is the bound of the local background. The inner window can be decided by the boundary pixels of the test superpixel, i.e., the superpixels neighboring the (centered) test superpixel are removed from the local background.

Construction of BCD: First, we use the dual-window strategy [53] to obtain the BCD of the current test superpixel, as shown in Fig. 2. Note that each superpixel has its local adjacent superpixels. According to Fig. 2, the local background superpixels of the test superpixel can be found according to the following three conditions.

1) They are within or on the outer square window with side length M.
2) They do not contain the superpixels that are closely neighboring the test superpixel.
3) They are not contaminated by potential abnormal superpixels.

Condition 1) is considered to reduce the size of the search area. Without loss of generality, we set $M = 5S$ to guarantee that the side length of the outer window is larger than the size of the test superpixels. When the test superpixel is only part of a large ship target, condition 2) guarantees that the set of background superpixels are still pure. Condition 3) is related to the heterogeneous sea clutter and crowded target scenarios, where some abnormal superpixels may exist in the local background. Here, an abnormal superpixel is defined as a superpixel with outlier intensities. For the pth test superpixel, the superpixels that satisfy conditions 1) and 2) are first found and then refined by removing outliers if their averaged intensities are outside the interval $[V_{p,\min}, V_{p,\max}]$, where

$$V_{p,\min} = \bar{\mu}_p - 2\bar{\sigma}_p \quad (10)$$

$$V_{p,\max} = \bar{\mu}_p + 2\bar{\sigma}_p \quad (11)$$

$\bar{\mu}_p$ and $\bar{\sigma}_p$ are, respectively, the mean value and the standard deviation of intensities of all the pixels contained by the superpixels that satisfy conditions 1) and 2). From (10) and (11), we have a 95% confidence interval to believe that the superpixels with average intensities in $[V_{p,\min}, V_{p,\max}]$ are not abnormal superpixels [54]. Let Θ_p denote the index set of the superpixels in the image that satisfy the above conditions 1)–3) for the pth test superpixel. The BCD of FVs for the

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pth test superpixel can be formulated as

$$\boldsymbol{\Psi}_p = \left[\mathbf{f}_i, \forall i \in \Theta_p\right] \in \mathbb{R}^{3N_G \times I_p} \tag{12}$$

where $I_p = |\Theta_p|$.

Calculation of SR Coefficients: Next, we calculate the SR vector for the FV $\mathbf{f}_p$ (of the pth test superpixel) on its BCD $\boldsymbol{\Psi}_p$ via NSLR

$$\hat{\boldsymbol{\theta}}_p = \arg\min_{\boldsymbol{\theta}_p} \|\mathbf{f}_p - \boldsymbol{\Psi}_p \boldsymbol{\theta}_p\|_2^2 + \lambda \|\mathbf{d}_p \odot \boldsymbol{\theta}_p\|_2^2,$$
$$\text{s.t.} \sum_{i=1}^{I_p} \theta_{p,i} = 1, \ \theta_{p,i} \geq 0, \ \forall i \in \{1, 2, \ldots, I_p\} \tag{13}$$

where $\boldsymbol{\theta}_p$ is the SR vector for the pth superpixel, λ is the regularization parameter, and $\odot$ represents the Hadamard product. In (13), $\mathbf{d}_p = [d_{p,1}, \ldots, d_{p,i}, \ldots, d_{p,I_p}] \in \mathbb{R}^{I_p \times 1}$ is the locality descriptor to measure the Euclidean distances between the FV $\mathbf{f}_p$ and each column vector in $\boldsymbol{\Psi}_p$, where

$$d_{p,i} = \exp\left(\|\mathbf{f}_p - \boldsymbol{\Psi}_{p,i}\|_2 / \nu\right) \tag{14}$$

$\nu > 0$ is a constant to represent the exponential decay speed, $\boldsymbol{\Psi}_{p,i}$ is the ith column of the matrix $\boldsymbol{\Psi}_p$, and $i \in \{1, 2, \ldots, I_p\}$. The locality descriptor $\mathbf{d}_p$ encourages that the FV $\mathbf{f}_p$ is represented by the FVs in BCD similar to itself. The constraint of $\sum_{i=1}^{I_p} \theta_{p,i} = 1$ in (13) emphasizes the shift-invariant feature of NSLR. The nonnegative constraint $\{\theta_{p,i} \geq 0, \ \forall i\}$ concerns the negative correlation characteristic between the target FVs and clutter FVs [28]. A target FV and a clutter FV cannot be represented by each other with nonnegative constraint $\{\theta_{p,i} \geq 0, \forall i\}$, which enhances the discrimination ability between the ship targets and the background sea clutter. Note that the commonly used l_0-norm sparsity constraint [37] is not presented in (13), since the locality descriptor $\mathbf{d}_p$ directly leads to sparsity [44] (but not necessarily vice versa).

The optimization problem in (13) is a quadratic programming one with inequality and equality constraints and can be solved by the existing optimization tools such as CVX [55], which iteratively updates the sparse solution of (13). To speed up the NSLR, a fast algorithm [46] (without iterative processing) is used here to obtain the approximate solution of (13). First, we calculate the analytical solution of (13) by ignoring the nonnegative constraint $\{\theta_{p,i} \geq 0, \ \forall i\}$

$$\hat{\boldsymbol{\theta}}_p = \left[\left(\boldsymbol{\Psi}_p^T - \mathbf{1}\mathbf{f}_p^T\right)\left(\boldsymbol{\Psi}_p^T - \mathbf{1}\mathbf{f}_p^T\right)^T + \lambda \text{diag}\left(\mathbf{d}_p \odot \mathbf{d}_p\right)\right]^{-1} \mathbf{1} \tag{15}$$

where $\mathbf{1} \in \mathbb{R}^{I_p \times 1}$ denotes the all-ones vector and diag represents the vector diagonalization operation. Then, a nonnegative constraint is imposed on $\hat{\boldsymbol{\theta}}_p$ [46]

$$\hat{\theta}_{p,i} \leftarrow \hat{\theta}_{p,i} + \left|\min\left\{0, \theta_p^*\right\}\right| \tag{16}$$

where $i = 1, 2, \ldots, I_p$, and

$$\theta_p^* = \min_i \left\{\hat{\theta}_{p,i}, i = 1, 2, \ldots, I_p\right\}. \tag{17}$$

After (17), the coefficients in $\hat{\boldsymbol{\theta}}_p$ smaller than the mean value of all the coefficients in $\hat{\boldsymbol{\theta}}_p$ are set to 0, to enhance the sparsity. Finally, $\hat{\boldsymbol{\theta}}_p$ is renormalized to satisfy the shift-invariant feature

$$\hat{\boldsymbol{\theta}}_p \leftarrow \hat{\boldsymbol{\theta}}_p / \sum_{i=1}^{I_p} \hat{\theta}_{p,i}. \tag{18}$$

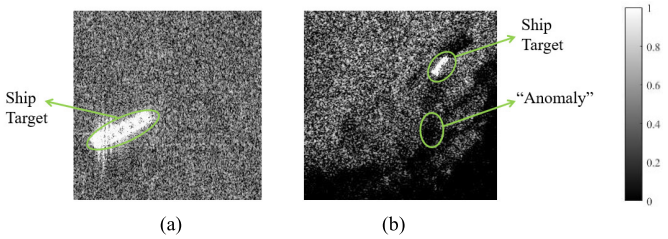


Fig. 3. (a) SAR image with homogeneous background clutter. (b) SAR image with heterogeneous background clutter.

Calculation of Representation Residuals: Based on the estimated SR vector $\hat{\boldsymbol{\theta}}_p$, the representation residual of NSLR is calculated as

$$r_p = \|\mathbf{f}_p - \boldsymbol{\Psi}_p \hat{\boldsymbol{\theta}}_p\|_2^2, \quad \text{for } p = 1, 2, \ldots, P \tag{19}$$

which can be used to distinguish between the ship targets and the sea clutter. If the test superpixel belongs to the sea clutter, its representation residual is close to 0 (since the clutter FV lies in the low-dimensional subspace spanned by the BCD). On the contrary, if the test superpixel is occupied by a ship target, the corresponding representation residual is a large value, because the target FV cannot be linearly represented by the clutter FVs in BCD. Therefore, if r_p is larger than a predefined decision threshold, the pth superpixel can be labeled as a ship target superpixel.

D. Step 4: Filtering: Embedding Target Features

The NSLR (introduced in Section II-C) can be regarded as a kind of "anomaly" detection method [39], [61], which is based on the differences between the ship targets and the background clutter and does not assume any *a priori* information about the ship targets. Accordingly, the NSLR in Section II-C is expected to work well in the presence of homogeneous background clutter, as in Fig. 3(a). In practice, SAR images sometimes are deteriorated by the heterogeneous sea clutter caused by current patterns, distances between the ship targets and land/islands, and the wakes of ships. As shown in Fig. 3(b), some anomaly areas (except for target areas) that exist in the SAR image with heterogeneous clutter may increase the number of false alarms. In this section, we design two filters based on FV to embed inherent knowledge of the ship targets into the NSLR. These two filters are helpful to enhance the robustness of NSLR to heterogeneous clutter.

In general, the ship targets exhibit the following two features in SAR images: 1) sparsity in spatiality. The ship targets occupy much fewer pixels than the sea clutter in SAR images and 2) larger intensity values. The ship targets often appear as bright parts of the SAR image. As shown in Fig. 4, the above *a priori* features of the ship targets can be reflected by the zero-order and first-order information in FVs. In Fig. 4(a), we show an SAR image chip and its superpixel segmentation results. In Fig. 4(b) and (c), the weights and mean values in the global GMM of Fig. 4(a) are depicted, where $N_G = 7$. The weight parameters $\{\omega_n, \forall n\}$ in GMM represent the ratios of pixels belonging to the corresponding Gaussian components in the SAR image (in spatiality). The mean values $\{\mu_n, \forall n\}$ are the average intensities of pixels in the corresponding Gaussian

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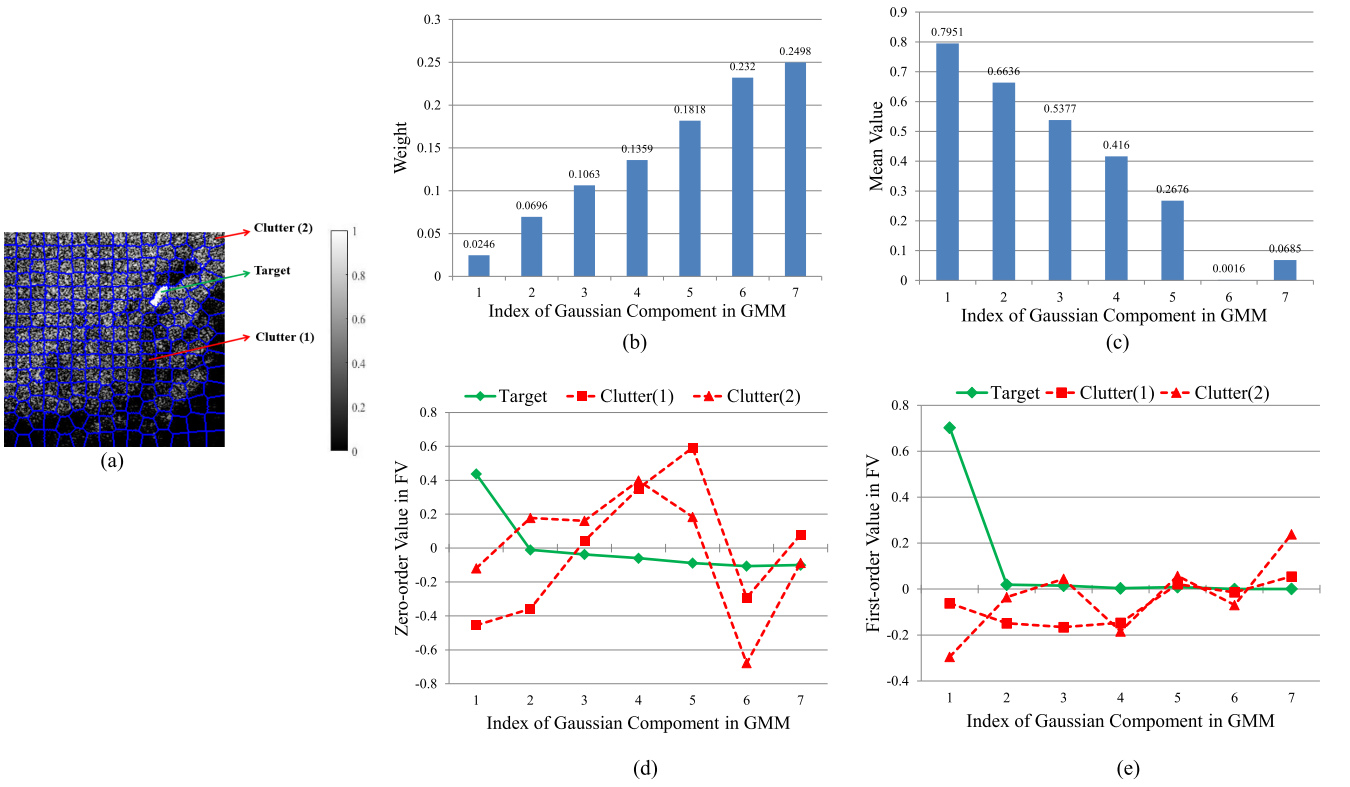


Fig. 4. Target features implied by the zero-order and first-order information in FVs. (a) Superpixel segmentation map of a Gaofen-3 SAR image. We use $N_G = 7$ Gaussian components to model the SAR image in (a). (b) Weight values $\{\omega_1, \omega_2, \ldots, \omega_7\}$ in the global GMM of (a). (c) Mean values $\{\mu_1, \mu_2, \ldots, \mu_7\}$ in the global GMM of (a). (d) Comparison of the zero-order values in the FVs of the selected target and clutter superpixels [see (a)]. (e) Comparison of the first-order values in the FVs of the selected target and clutter superpixels.

components. It can be observed from Fig. 4(b) and (c) that the Gaussian component with the smallest weight has the largest mean value (see the first component). This component is consistent with *a priori* features of ship targets, that is: 1) sparsity in spatiality and 2) large intensity values. Therefore, it is expected that the ship targets belong to the Gaussian component with the smallest weight and the largest mean value, while the background clutter tends to occupy the other components. From (4), (6), and (7), we can see that zero-order and first-order values in FV represent the "scores" of the current superpixel on the weight and mean value parameters in the global GMM. In Fig. 4(d) and (e), the zero-order and first-order values in the FVs of the target and clutter superpixels marked in Fig. 4(a) are provided. Fig. 4(d) and (e) show that the target superpixel has the highest scores on the smallest weight and the largest mean value corresponding to the first Gaussian component. On the contrary, clutter superpixels show smaller scores on the parameters of the first Gaussian component and have larger scores on those of other Gaussian components.

According to the above observations, we design two filters to enhance the ship targets and suppress the background sea clutter

$$\text{zero-order filter: } \xi_p^{(0)} = \sum_{n=1}^{N_G} f_{p,\omega_n}/\omega_n \tag{20}$$

$$\text{first-order filter: } \xi_p^{(1)} = \sum_{n=1}^{N_G} f_{p,\mu_n} \times \mu_n \tag{21}$$

for $p = 1, 2, \ldots, P$. The values of $\xi_p^{(0)}$ and $\xi_p^{(1)}$ can be regarded as the (weighted-) sum of the zero-order and first-order elements in FV of the pth superpixel, where the zero-order and first-order elements are weighted by $1/\omega_n$ and μ_n, respectively. $\xi_p^{(0)}$ and $\xi_p^{(1)}$ tend to be large values if the pth superpixel is the target superpixel since the target superpixel has large scores on the smallest weight and the largest mean value in GMM. On the other hand, clutter superpixels share the small values of $\xi_p^{(0)}$ and $\xi_p^{(1)}$ due to their larger scores on the large weights and the small mean values in GMM. Table I shows the values of $\xi_p^{(0)}$ and $\xi_p^{(1)}$ of the marked target and clutter superpixels in Fig. 4(a), confirming the effectiveness of the filters in (20) and (21) in the tasks of target enhancement and clutter suppression.

Next, the performance of NSLR representation residual r_p in (19) is enhanced using filters in (20) and (21)

$$R_p = r_p \times \tilde{\xi}_p^{(0)} \times \tilde{\xi}_p^{(1)} \tag{22}$$

where

$$\tilde{\xi}_p^{(0)} = \frac{\xi_p^{(0)} - \min_p\left\{\xi_p^{(0)}, \forall p\right\}}{\max_p\left\{\xi_p^{(0)}, \forall p\right\} - \min_p\left\{\xi_p^{(0)}, \forall p\right\}}$$

$$\tilde{\xi}_p^{(1)} = \frac{\xi_p^{(1)} - \min_p\left\{\xi_p^{(1)}, \forall p\right\}}{\max_p\left\{\xi_p^{(1)}, \forall p\right\} - \min_p\left\{\xi_p^{(1)}, \forall p\right\}}$$

are the normalized values (located in [0, 1]) of $\xi_p^{(0)}$ and $\xi_p^{(1)}$, respectively, and $p = 1, 2, \ldots, P$. The enhanced representation residual R_p in (22) is more robust to the heterogeneous

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TABLE I
VALUES OF FILTERS FOR THE SELECTED SUPERPIXELS IN FIG. 4(A)

	Zero-order: $\xi_p^{(0)}$	First-order: $\xi_p^{(1)}$
Target	**15.4806**	**0.5832**
Clutter (1)	-17.2634	-0.1104
Clutter (2)	-0.1189	-0.279

sea clutter than the original representation residual in (19), thanks to the embedded target features.

E. Step 5: Thresholding Operation

Global thresholding can be performed on the map of enhanced representation residuals $\{R_p, \forall p\}$ to obtain the binary detection results $\{\gamma_p, \forall p\}$

$$\begin{cases} \gamma_p = 1, & \text{if } R_p \geq \eta \\ \gamma_p = 0, & \text{if } R_p < \eta \end{cases} \tag{23}$$

where the global threshold η is adaptively determined using the classical two-parameter method [57]

$$\eta = \mu^{(R)} + \zeta \sigma^{(R)}$$

$\mu^{(R)}$ and $\sigma^{(R)}$ denote the mean value and the standard deviation of $\{R_p, p = 1, 2, \ldots, P\}$, respectively, and ζ represents the adjustment factor. In practice, the adjustment factor ζ is suggested to range from 3 to 14 [58].

The implementation of the proposed NSLR detector is summarized in Algorithm 1.

Algorithm 1 NSLR-Based Detector

Input: Test SAR image.
Step 1: Superpixel segmentation for the SAR image by using the steps 1a-1e in Section II.A.
Step-2: Calculation of the FV for each superpixel in the SAR image via (5)-(9).
Step-3: Construction of the BCD of FVs for each superpixel based on Fig. 2. Calculation of the NSLR representation residual for each FV on its corresponding BCD, according to (15)-(19).
Step-4: Filter the representation residual map via (22).
Step-5: Obtain the binary detection results based on (23).
Output: Binary detection results.

III. EXPERIMENTAL RESULTS

In this section, we conduct experiments on Gaofen-3 SAR images to demonstrate the effectiveness of the proposed NSLR detector. Here, the Gaofen-3 SAR images are collected with different image resolutions in homogeneous/heterogeneous sea clutter scenarios [56], [59]. The parameters of the data set used in this section are provided in Table II. The ground truth of the ship targets in SAR images (designed or confirmed by technical experts) can be found in open data sets [56], [59]. In this section, the hyper-parameter settings for the proposed and other competitive detectors are first discussed. Then,

TABLE II
PARAMETERS OF SAR IMAGES USED FOR EXPERIMENTS[2]

	Fig. 5(a), Fig. 6(a), and SAR images for Table VI	Fig. 7(a) and Fig. 8(a)	Fig. 9(a)	Fig. 10(a)
Satellite	Gaofen-3	Gaofen-3	Gaofen-3	Gaofen-3
Polarization	Single (HH)	Single (HH)	Single	Single
Frequency Band	C	C	C	C
Incidence Angle	20°-50°	20°-41°	-	-
Resolution	3m×3m	8m×8m	3m×3m	1m×1m
Imaging Mode	UFS	QPSI	-	-
Size	256×256	256×256	3000×3000	3000×2477
Sea State Code	-	-	3	4

TABLE III
CF (IN dB) OF THE PROPOSED NSLR DETECTOR WITH DIFFERENT VALUES OF λ IN (13) AND ν IN (14)

	$\lambda = 0.5$	$\lambda = 1$	$\lambda = 10$	$\lambda = 15$
$\nu = 0.5$	13.2068	12.0384	11.9896	13.1521
$\nu = 1$	12.4215	13.3719	12.2653	13.1408
$\nu = 10$	11.3176	12.1483	11.1309	11.0667
$\nu = 15$	12.7256	9.8838	11.8914	11.7789

we compare the detection performance and complexity of the superpixel-based local contrast measure (SLCM) [29], local contrast of FV (LCFV) [9], superpixel-level statistical dissimilarity (SSD) [31], and the proposed NSLR detector.

The performance of the above detectors is evaluated in terms of the contrast factor (CF) [60], probability of detection ($\mathrm{Pr_d}$), probability of false alarm ($\mathrm{Pr_{fa}}$), receiver operating characteristic curve (ROC), and area under the curve (AUC). CF, in essence, is defined as the ratio of the test statistic of the "weakest" target superpixel to the average test statistic of the clutter superpixels (before the global thresholding operation). In this article, CF is calculated as

$$\mathrm{CF} \triangleq 10\log_{10} \frac{\min_w \left\{ R'_w, w = 1, 2, \ldots, N_{\mathrm{Truetar}} \right\}}{\frac{1}{|\mathrm{Clu}|} \sum_{p \in \mathrm{Clu}} R_p} \tag{24}$$

where N_{Truetar} represents the number of ship targets in the SAR image, $R'_w = \max_p \{R_p, p \in \mathrm{Tar}^{(w)}\}$, "$\mathrm{Tar}^{(w)}$" and "Clu" are the ground truth of the wth target and the background clutter area in the SAR image, respectively, and |Clu| denotes the number of superpixels in the background clutter. Note that the definition of CF is different from that of SCR. SCR reflects the difference between the ship targets and the clutter in the original SAR image, while CF describes that in the SAR image after the target enhancement via designed test statistics. A larger value of CF indicates that it is easier to detect weak ship targets from the strong background clutter by thresholding

[2]In Table II, "-" denotes the default values which are not provided in original data sets [56], [59].

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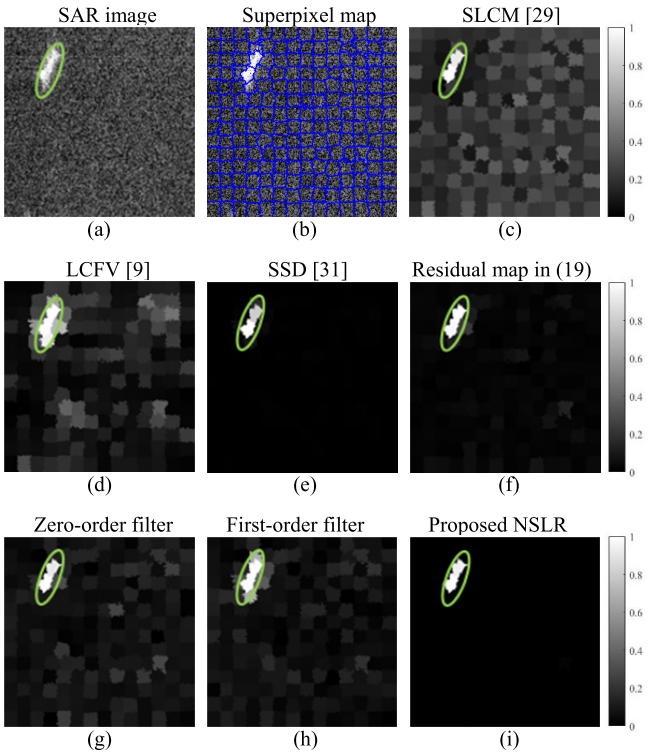


Fig. 5. (a) SAR image with homogeneous background clutter (high SCR). (b) Superpixel segmentation result, where $S = 18$, $\tau = 0.4$. Test statistic maps (before thresholding) of (c) SLCM [29], (d) LCFV [9], and (e) SSD [31]. (f) Representation residual map $\{r_p,\ p = 1, 2, \ldots, P\}$ in (19) without zero-order and first-order filters. (g) Map of the zero-order filter in (20). (h) Map of the first-order filter in (21). (i) Enhanced residual map $\{R_p,\ p = 1, 2, \ldots, P\}$ in (22).

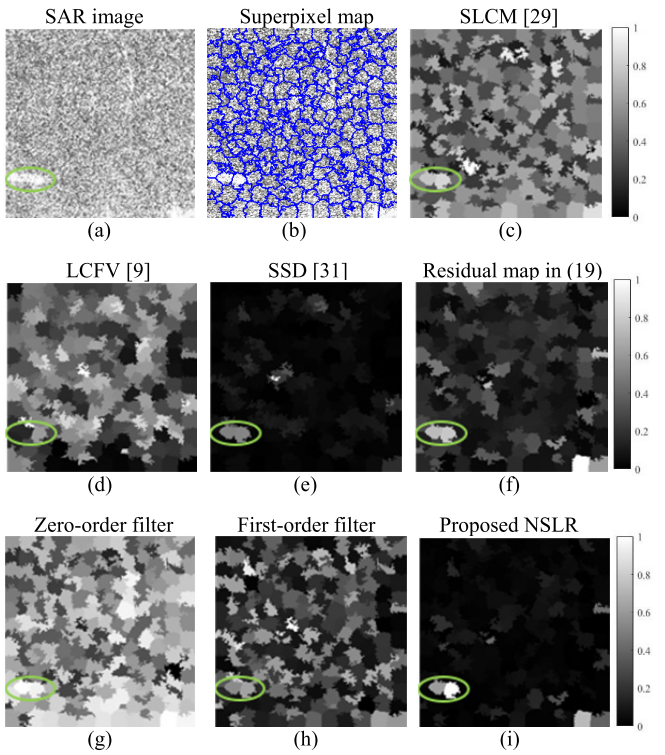


Fig. 6. (a) SAR image with homogeneous background clutter (low SCR). (b) Superpixel segmentation result, where $S = 18$, $\tau = 0.4$. Test statistic maps (before thresholding) of (c) SLCM [29], (d) LCFV [9], and (e) SSD [31]. (f) Representation residual map $\{r_p,\ p = 1, 2, \ldots, P\}$ in (19) without zero-order and first-order filters. (g) Map of the zero-order filter in (20). (h) Map of the first-order filter in (21). (i) Enhanced residual map $\{R_p,\ p = 1, 2, \ldots, P\}$ in (22).

test statistics. $\mathrm{Pr_d}$ and $\mathrm{Pr_{fa}}$ in ROCs are defined as

$$\mathrm{Pr_d} = N_{\mathrm{Dectar}}/N_{\mathrm{Truetar}} \tag{25}$$

$$\mathrm{Pr_{fa}} = N_{\mathrm{Decfal}}/(P - N_{\mathrm{Truetar}}) \tag{26}$$

where N_{Dectar} and N_{Decfal} denote the number of detected targets and false alarm superpixels, respectively.

A. Hyper-Parameter Settings

The proposed and the other competitive superpixel-based methods require several hyper-parameters to be operated. At the time of superpixel generation via SLIC, a large value of superpixel size S is helpful to accelerate the detection task but may degrade the detection performance. This is because a very large superpixel may contain both the target and clutter pixels. The superpixel size S can be selected in accordance with a fixed percentage (e.g., 25%–100%) of the average number of pixels in the area occupied by a ship target. This setting maintains the shape of the ship targets well. For example, the ship target in Fig. 5 occupies approximately 600 pixels with $S = 18$ ($S^2/600 \approx 50\%$). The above setting for superpixel size also significantly reduces the complexity of detection when compared with that of pixel-level detection [25]. In Sections III-B and III-C, the robustness of our proposed method to the superpixel size S and the target size will be investigated. The balancing parameter τ in SLIC is set to be a value smaller than (or equal to) 1, to achieve a good equilibrium between the intensity and spatiality features, as suggested in [9] and [28]. Achanta *et al.* [25] show that only a few iterations are sufficient for SLIC to generate compact superpixels. In this article, the maximum iteration count of SLIC is fixed to $Iter = 20$. A large value of N_{G}, that is, the number of Gaussian components in GMM, increases the complexity of parameter estimation for GMM, while a small value of N_{G} (e.g., $N_{\mathrm{G}} \leq 2$) cannot effectively model the SAR image due to the heterogeneous background clutter. In [9], [11], and [28], $N_{\mathrm{G}} = 7$ is suggested for a better compromise between complexity and performance and is also adopted in this article. From (13), we can observe that the regularization parameter λ balances the relative importance between the representation residual and the locality. In (14), the locality is further emphasized with a small value of exponential decay speed ν. In Table III, we show the CF performance of the proposed NSLR detector in Algorithm 1 using the test image in Fig. 4(a), where $\lambda = \{0.5, 1, 10, 15\}$ and $\nu = \{0.5, 1, 10, 15\}$, respectively. We can see that NSLR provides high CF values (CF $>$ 10 dB) with most pairs of (λ, ν). Without loss of generality, we set $\nu = \lambda = 10$ in the experiments hereinafter.

B. Anticlutter Performance: CFs

In this section, we compare the anticlutter performance of SLCM [29], LCFV [9], SSD [31], and the proposed NSLR method in terms of CF.

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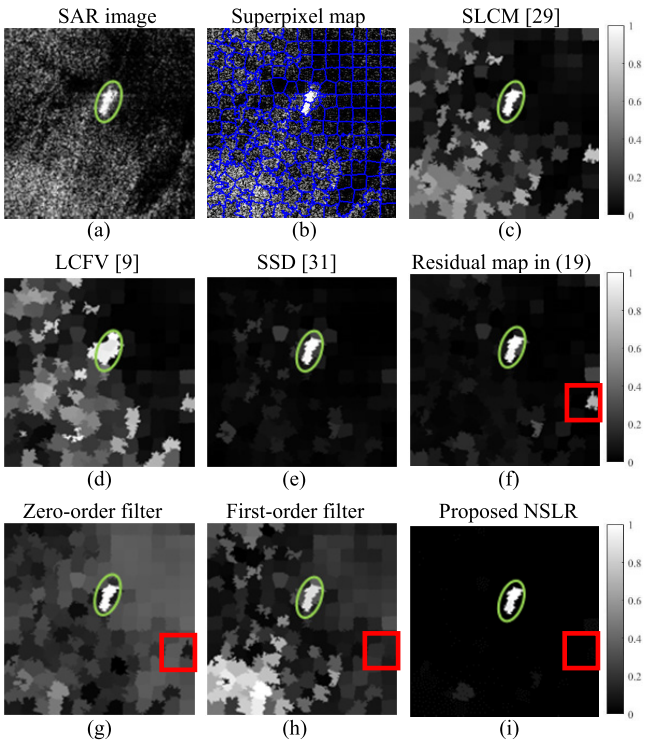


Fig. 7. (a) SAR image with heterogeneous background clutter (single target scenario). (b) Superpixel segmentation result, where $S = 18$, $\tau = 0.4$. Test statistic maps (before thresholding) of (c) SLCM [29], (d) LCFV [9], and (e) SSD [31]. (f) Representation residual map $\{r_p,\ p = 1, 2, \ldots, P\}$ in (19) without zero-order and first-order filters. (g) Map of the zero-order filter in (20). (h) Map of the first-order filter in (21). (i) Enhanced residual map $\{R_p,\ p = 1, 2, \ldots, P\}$ in (22).

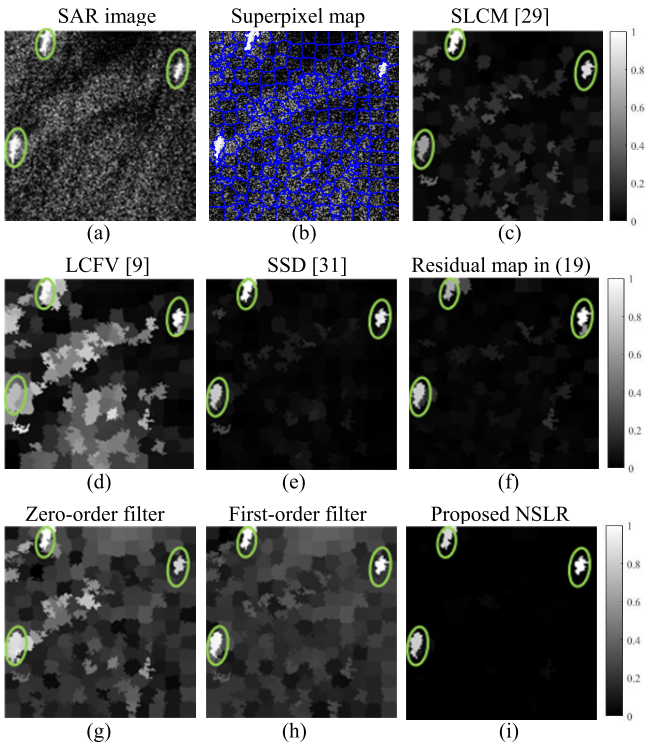


Fig. 8. (a) SAR image with heterogeneous background clutter (multitarget scenario). (b) Superpixel segmentation result, where $S = 18$, $\tau = 0.4$. Test statistic maps (before thresholding) of (c) SLCM [29], (d) LCFV [9], and (e) SSD [31]. (f) Representation residual map $\{r_p,\ p = 1, 2, \ldots, P\}$ in (19) without zero-order and first-order filters. (g) Map of the zero-order filter in (20). (h) Map of the first-order filter in (21). (i) Enhanced residual map $\{R_p,\ p = 1, 2, \ldots, P\}$ in (22).

In Figs. 5–8, we show the test statistic maps of the aforementioned methods in scenarios with homogeneous background clutter (in high/low SCRs) and heterogeneous background clutter (with one target and multitargets), respectively. Ship targets are marked using green ellipses. All the test statistic maps are normalized in the range [0, 1]. Superpixel segmentation maps are illustrated in Figs. 5(b)–8(b), in which SLIC [25] provides the compact target and clutter superpixels. From Fig. 5, we can see that LCFV [9] and SSD [31] perform better in terms of target enhancement and clutter suppression than SLCM [29]. This is because LCFV [9] and SSD [31] exploit better the statistical differences between the target and clutter superpixels than SLCM [29]. In Fig. 5(f), we can see that the representation residual map of NSLR provides a clear background, thanks to the exploitation of subspace features of FVs. As observed in Fig. 5(f)–(i), filters proposed in (20) and (21) improve the discrimination ability between the target and the sea clutter. Fig. 6 shows that the proposed NSLR can maintain its superiority even with strong background clutter interference in the original SAR image. Figs. 7 and 8 illustrate the test statistic maps of different ship detectors in scenarios with heterogeneous background clutter. SLCM [29] and SSD [31] show better target enhancement performance than LCFV [9] since LCFV [9] only focuses on the anomaly areas in SAR images and heterogeneous clutter may deteriorate the performance of anomaly detection. In Figs. 7 and 8, the proposed NSLR detector still generates the test statistic maps with higher contrast between the target and clutter superpixels. Especially, it is observed from Fig. 7(g) and (h) that the zero-order and first-order filters are helpful to suppress the sea clutter and relatively enhance the ship targets [see the clutter areas circled by red rectangles in Fig. 7(f)–(i)].

TABLE IV
CF (IN dB) OF FIGS. 5–8

	CF			
	Fig. 5	Fig.6	Fig. 7	Fig. 8
SLCM [29]	6.7028	2.9227	7.3817	10.8640
LCFV [9]	8.3033	4.1389	6.4049	5.8475
SSD [31]	24.3281	9.9300	13.9098	13.2170
Proposed NSLR	**33.5088**	**12.0137**	**24.5675**	**24.6643**

Quantitatively, higher CFs of Figs. 5–8 are provided by the proposed NSLR detector, as shown in Table IV. This implies that our NSLR achieves better anticlutter performance. Again, these quantitative results confirm the effectiveness of the proposed NSLR detector.

C. Detection Performance

In this section, we show the ROCs of the proposed NSLR detector and other competitive detectors using two Gaofen-3 SAR images, both of which cover large-scale scenarios.

An SAR image with 3000 × 3000 pixels and its superpixel segmentation map are provided in Fig. 9(a) and (b), respectively. This SAR image contains six multiscale ship targets. The ship targets at the upper left corner and lower

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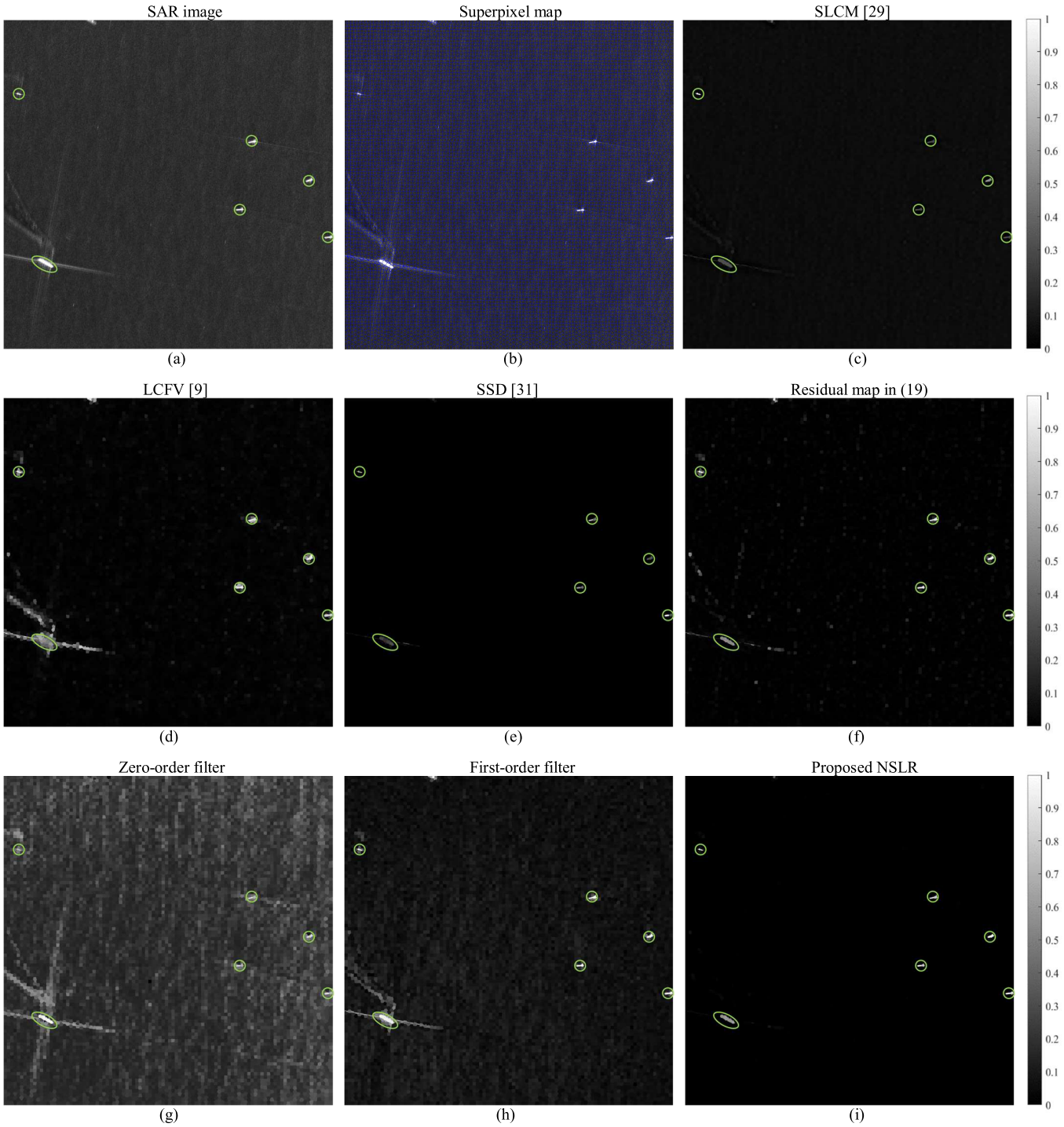


Fig. 9. (a) SAR image containing six multiscale ship targets. (b) Superpixel segmentation result, where $S = 30$, $\tau = 0.3$. Test statistic maps (before thresholding) of (c) SLCM [29], (d) LCFV [9], and (e) SSD [31]. (f) Representation residual map $\{r_p,\ p = 1, 2, \ldots, P\}$ in (19) without zero-order and first-order filters. (g) Map of the zero-order filter in (20). (h) Map of the first-order filter in (21). (i) Enhanced residual map $\{R_p,\ p = 1, 2, \ldots, P\}$ in (22).

left corner of Fig. 9(a) occupy approximately 500 and 3500 pixels, respectively. The superpixel size is $S = 30$ in Fig. 9(a), where $3500 > S^2 > 500$. From Fig. 9(b), we can see that the setting of $S = 30$ works well for maintaining the shape of both large and small ship targets. In Fig. 9(c)–(i), we show the test statistic maps of SLCM [29], LCFV [9], SSD [31], and the proposed NSLR detectors before the thresholding operation. From Fig. 9, we can see that LCFV [9] and NSLR (without filters) accurately locate all the ship targets but LCFV fails to suppress the scattering tails. SLCM [29] and SSD [31] generate a clear background but show weak amplitudes of some small ship targets. The proposed NSLR detector with zero-order and first-order filters obtains enhanced ship targets and clearer background simultaneously.

Fig. 10(a) illustrates an SAR image (3000 × 2477 pixels) with congested ship targets and heterogeneous background clutter. As shown in Fig. 10, SLCM [29] and LCFV [9] provide low contrast between the ship targets and the sea clutter. SSD

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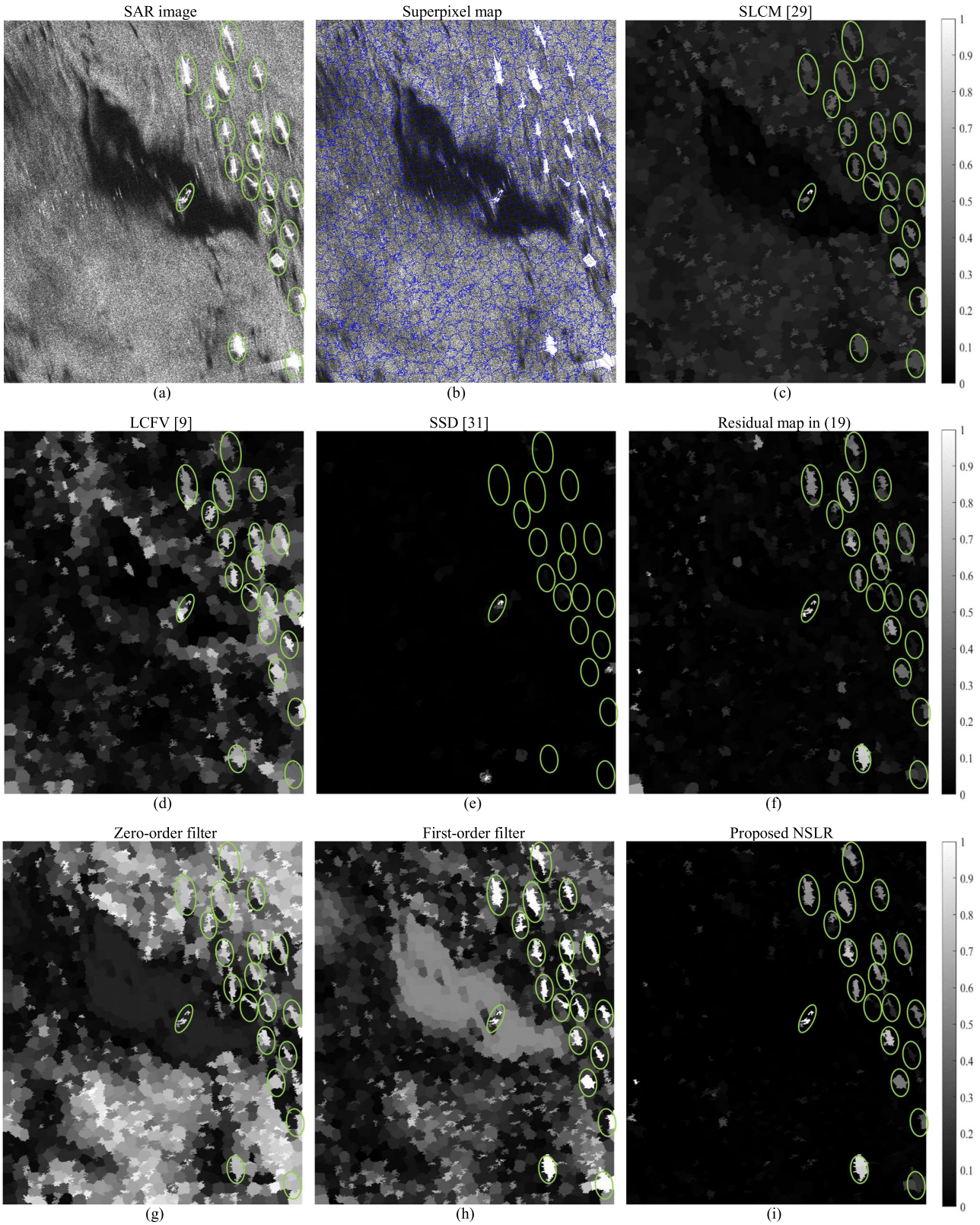


Fig. 10. (a) SAR image with congested ship targets and heterogeneous background clutter. (b) Superpixel segmentation result, where $S = 70$, $\tau = 0.3$. Test statistic maps (before thresholding) of (c) SLCM [29], (d) LCFV [9], and (e) SSD [31]. (f) Representation residual map $\{r_p,\ p = 1, 2, \ldots, P\}$ in (19) without zero-order and first-order filters. (g) Map of zero-order filter in (20). (h) Map of first-order filter in (21). (i) Enhanced residual map $\{R_p,\ p = 1, 2, \ldots, P\}$ in (22).

fails to locate most ship targets. This is because SSD strictly relies on the assumption that the SAR image follows a Gamma distribution and adopts the corresponding parameter estimation. The heterogeneous background clutter in Fig. 10 may degrade the performance of parameter estimation for Gamma distribution and further detection performance. Comparing

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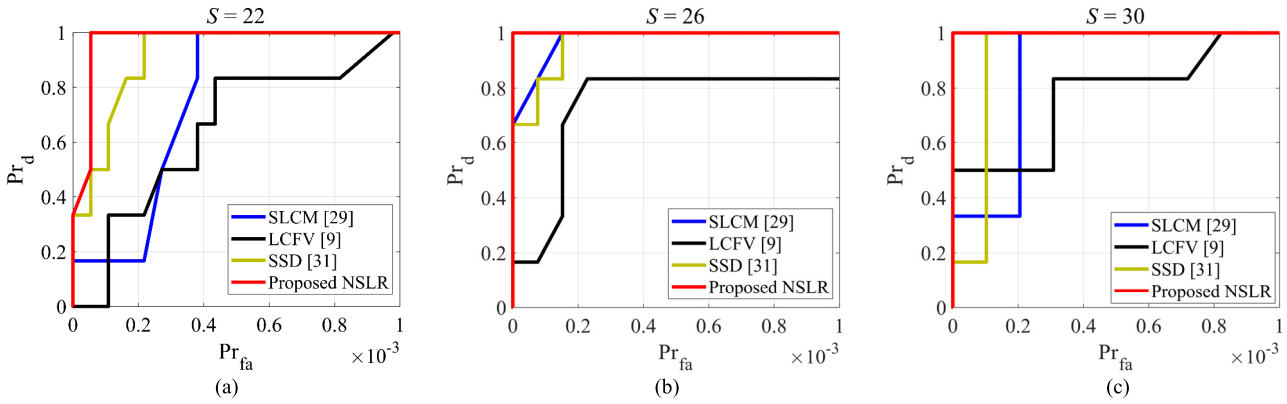


Fig. 11. ROC curves of Fig. 9 with different values of the superpixel size S, where $\tau = 0.3$. (a) $S = 22$. (b) $S = 26$. (c) $S = 30$.

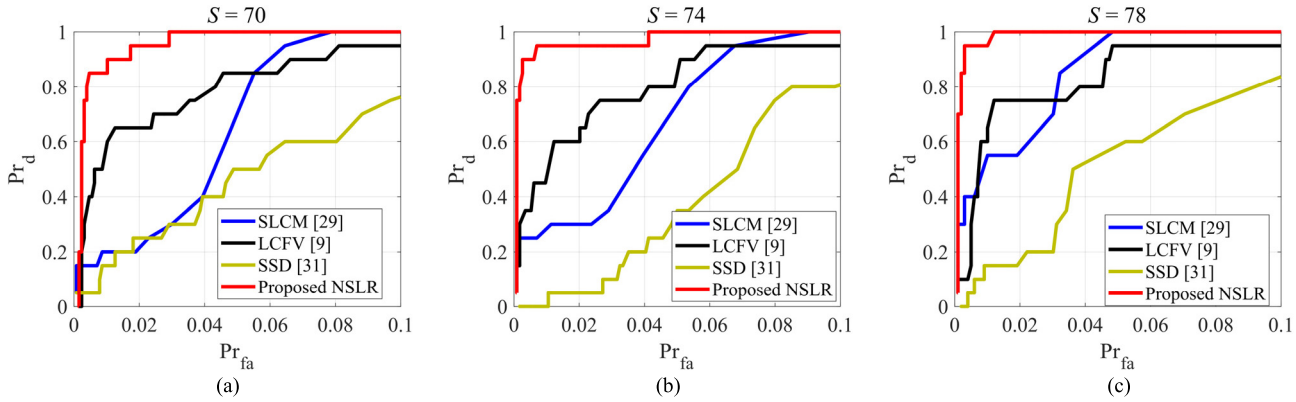


Fig. 12. ROC curves of Fig. 10 with different values of the superpixel size S, where $\tau = 0.3$. (a) $S = 70$. (b) $S = 74$. (c) $S = 78$.

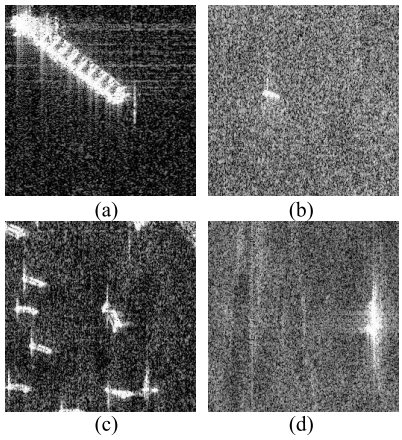


Fig. 13. Four examples in the SAR image data set for Fig. 14. (a) SAR image with a large ship target. (b) SAR image with a small ship target. (c) SAR image with multiple small ship targets. (d) SAR image with the heterogeneous sea clutter.

TABLE V
AUC VALUES OF ROCS IN FIGS. 11 AND 12 WITH DIFFERENT SUPERPIXEL SIZES

	Fig. 11			Fig. 12		
	$S = 22$	$S = 26$	$S = 30$	$S = 70$	$S = 74$	$S = 78$
SLCM [29]	0.9997	≈1	0.9999	0.9628	0.9665	0.9838
LCFV [9]	0.9996	0.9999	0.9994	0.9667	0.9707	0.9677
SSD [31]	0.9999	≈1	0.9999	0.9165	0.9238	0.9355
Proposed NSLR	**≈1**	**≈1**	**≈1**	**0.9950**	**0.9966**	**0.9981**

Fig. 10(c)–(e) with Fig. 10(f), we can observe that the subspace features of FVs exploited by NSLR are helpful in reducing the clutter level and enhancing the ship targets. Furthermore, the comparison between Fig. 10(f) and Fig. 10(i) shows that the proposed zero-order and first-order filters improve the ship detection performance.

SLCM [29], LCFV [9], SSD [31], and the proposed NSLR detector are based on the global thresholding of the test statistic maps. In Figs. 11 and 12, we plot the ROC curves of Figs. 9 and 10 versus the superpixel size S to thoroughly investigate the performance of ship detection, where $\mathrm{Pr_d}$ and $\mathrm{Pr_{fa}}$ in ROCs are calculated according to (25) and (26) and different global thresholds. We can observe from Figs. 11 and 12 that the proposed NSLR detector achieves better global detection performance than other competitive detectors with different superpixel sizes. The AUCs of ROCs in Figs. 11 and 12 are shown in Table V. Higher AUCs are provided by NSLR especially for Fig. 12.

Next, we compare the aforementioned ship detection methods using a large data set with 1723 Gaofen-3 images [56] in terms of $\mathrm{Pr_d}$ and $\mathrm{Pr_{fa}}$. Some examples in this data set are shown in Fig. 13, which contains ship targets with different sizes and different sea clutter environments caused by different

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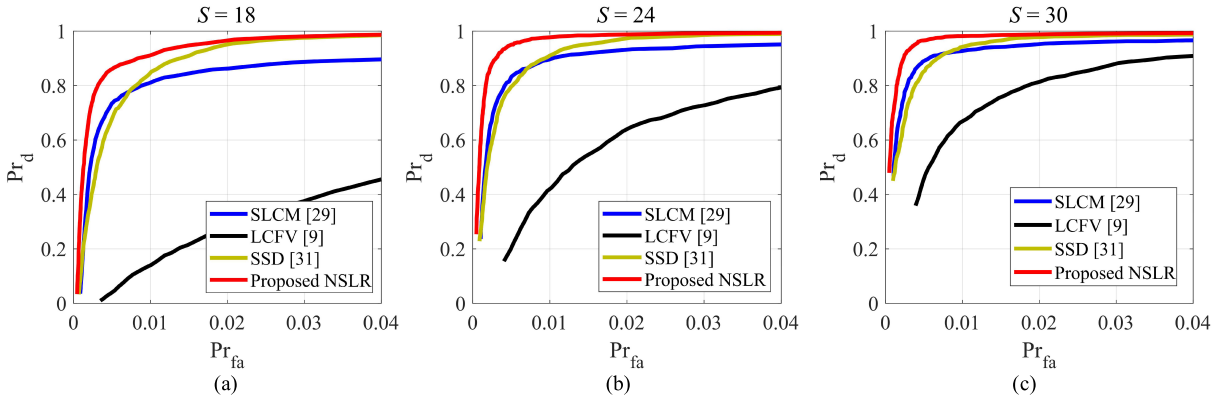


Fig. 14. ROC curves based on 1723 SAR images in [56] with different values of the superpixel size S, where $\tau = 0.4$. (a) $S = 18$. (b) $S = 24$. (c) $S = 30$.

TABLE VI
COMPARISON OF THE RUNNING TIME (S)

	Scene in Fig. 9 (S=30)	Scene in Fig. 10 (S=80)
SLCM [29]	428	465
LCFV [9]	570	566
SSD [31]	518	498
Proposed NSLR	1152	824

imaging conditions (e.g., incident angle, sea state). There are 1935 ship targets in this large data set. According to [56], the percentages of ship targets (in this data set) that their bounding boxes occupy less than 2.5%, 5%, and 10% pixels in a 256 × 256 SAR image chip are 31%, 74%, and 92%, respectively. Therefore, the sizes of most of the ship targets in this data set are relatively small. The smallest ship target occupies only 0.6% pixels in a 256 × 256 image chip. Here, we do not show the scenario with the smallest ship target in this data set and its detection results since they are highly similar to Fig. 8. In Fig. 14, we provide the ROC performance of SLCM [29], LCFV [9], SSD [31], and the proposed NSLR based on this data set containing 1723 Gaofen-3 SAR image chips. In Fig. 14(a)–(c), the percentages of pixels occupied by a superpixel in an image chip are approximately 0.4%, 0.8%, and 1.3% with the settings of $S = 18$, 24, and 30, respectively. Therefore, the setting of $S = 18$ in Fig. 14(a) can cover even the smallest ship target in this data set and the settings of $S = 24$ and 30 associate with slightly larger ship targets. As shown in Fig. 14, the proposed NSLR method consistently achieves better detection performance than other methods. Fig. 14 also shows that the performance of the proposed NSLR method is robust to the mild change in the superpixel size S.

D. Running Time

In Table VI, we compare the running times of SLCM [29], LCFV [9], SSD [31], and the proposed NSLR detector. MATLAB 2019a is used as the software platform, which is in the 64-bit Windows system with the Inter Core i7-8550U CPU and 8-GB RAM. From Table VI, we find that SLCM [29] has the minimum running time since it only requires the average intensities of superpixels. Compared with other detectors, the proposed NSLR detector slightly increases the computational time. This is mainly caused by the calculation of FVs and the operation of matrix inversion in (15). The FVs calculated in NSLR (and in LCFV [9]) can also be used for the subsequent target classification task [33], [34] and save the computational time therein. To achieve fast detection for SAR images with a larger size, accelerated matrix inversion techniques [44], [61], such as the elaborate FPGA and approximation algorithms, can be used for the matrix inversion in (15).

IV. CONCLUSION

In this article, we proposed a new NSLR detector for superpixel-based ship detection in SAR images, where FV-based subspace features are innovatively considered. The proposed NSLR detector is designed based on the assumption that the FV of a superpixel can be sparsely represented by the BCD constructed from clutter superpixels only under a null hypothesis. The representation residuals of FVs via NSLR are calculated to discriminate the ship targets and the background sea clutter. In addition, two filters based on FVs exploiting the intensity and spatiality features of the ship targets are developed to enhance the performance of the proposed NSLR method in heterogeneous sea clutter scenarios. Experiments based on a variety of Gaofen-3 SAR images demonstrate that compared with the existing and commonly used ship detectors, the proposed NSLR detector dramatically improves the detection performance with affordable increase of complexity. The exploiting of higher order information in SAR images and the combination of NSLR with ship target classification algorithms will be pursued in our future research.

ACKNOWLEDGMENT

The authors would like to thank the editors and anonymous reviewers for their valuable suggestions improving the quality of this article.

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